

1 April 25th, 2020 Meeting

1.1 Our Paper

Nick's Question 1.1 I think we can strengthen Theorem 3.2 from our paper for when we have a L^2_ρ initial condition. The result from Theorem 3.2 is that if we start with a SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

and then consider

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2}\Delta_x v(t, x) = b(x, v(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ v(0, x) = u(t_0, x) \end{cases}$$

then this new SPDE satisfies

$$\sup_{t \geq 0} \mathbb{E}|v(t, \cdot)|_\rho^2 < \infty.$$

But $v(t, x) = u(t + t_0, x)$ so really what this theorem is saying is that

$$\sup_{t \geq t_0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty.$$

But I was thinking about what happens for $t \in [0, t_0]$. From your paper with Huang [15], you proved that

$$(t, x) \mapsto E(|u(t, x)|^2)$$

is continuous on $(0, \infty) \times \mathbb{R}^d$ if assuming the appropriate conditions listed in [15]. But I think that this can imply that

$$t \mapsto E(|u(t, \cdot)|_\rho^2)$$

is continuous on $(0, \infty)$ and this is sort of what Theorem 2.3 from the Tessitore paper is saying [39].

Now, are there situations where we can say that

$$t \mapsto E(|u(t, \cdot)|_\rho^2)$$

is continuous on $[0, \infty)$? This can't be the case if we have δ_0 as the initial condition because

$$E(|u(0, \cdot)|_\rho^2) = E(|\delta_0(\cdot)|_\rho^2)$$

does not exist. But I believe that if we have a $\varphi \in L^2_\rho$ initial condition, then

$$t \mapsto E(|u(t, \cdot)|_\rho^2)$$

is continuous on $[0, \infty)$. More specifically, this would imply that

$$\sup_{0 \leq t \leq t_0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty$$

which further implies that

$$\sup_{t \geq 0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty.$$

This would be an improvement to the result we already proved.

1.2 Misiat and Stanzhytski Paper

Remark 1.1. A somewhat stonger result has already been proven in Misiats and Stanzhytski paper [30]. In their paper they consider the SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - Au(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in G, t > 0 \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

where A is any elliptic operator and $G \subseteq \mathbb{R}^d$. They proved that when starting with a L^2 function, hence not any L^2_ρ function, then

$$\sup_{t \geq 0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty.$$

However, in their paper they require that there exists a function $\psi \in L^1 \cap L^\infty$ and constant $\sigma_0 > 0$ such that

1. $|b(x, u)| \leq \sigma_0$
2. $|f(x, u)| \leq \psi(x)$ for any $(x, u) \in \mathbb{R}^d \times \mathbb{R}$

I don't believe that we required the first assumption to get the result but we only considered $f \equiv 0$, so if we do not assume that then I am not sure if we will also need the second assumption or not. However our proof relies on the Moment estimates from [15] which only apply to the case of $A = \frac{1}{2}\Delta$. Also the moment bounds are for the case for $f \equiv 0$ so extra work is going to need to be done if we are going to want to relax the second condition above which is used in [30]. However, since

$$\int_0^t \int_{\mathbb{R}^d} f(s, u(s, y))G(t-s, x-y) \, dx dy$$

is not a stochastic integral then we could copy the proof in [30] which deal with this integral exactly and keep the second condition listed above. This may help us to prove the existence of an invariant with non-zero f term with the δ_0 initial condition.

Nick's Question 1.2 In Theorem 1 in the paper [30]. They are trying to show that

$$E \left| \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(y, u(s, y)) dy dW(s, y) \right| < \text{Const.}$$

Here is a part of what they did:

$$\int_0^{t-1} \int_{\mathbb{R}^d} G^2(t-s, y) \, dy ds \leq C_1 \int_0^{t-1} \frac{1}{(t-s)^{d/2}} \, ds \leq C_2 < \infty$$

I don't see why the first inequality is true unless we are dealing with heat equation for which it would then be true since $G(t, x)$ would be the standard heat kernel

$$\begin{aligned} \int_0^{t-1} \int_{\mathbb{R}^d} G^2(t-s, y) \, dy ds &= \int_0^{t-1} G(2(t-s), 0) \, ds \\ &\leq \int_0^{t-1} \frac{1}{(t-s)^{d/2}} \, ds \end{aligned}$$

Also Misiat and Stanzhytski never make mention of any restriction on the correlation function of the Gaussian noise. What they are essentially doing is finding a bound on the second moment but for this to be done, we need to make assumptions on the correlation

function of the Gaussian noise, namely we need to assume

$$\int_{\mathbb{R}^d} \frac{f(d\xi)}{|\xi|^2} < \infty$$

Remark 1.2 (Uniqueness of the invariant measure). There is a property called exponential stability defined in [30]: A solution of an SPDE u is said to be exponentially stable if there exists $K > 0$ and $\gamma > 0$ such that for any t_0 and any other solution η with $\eta(t_0) \in \mathcal{F}_{t_0}$ and $\mathbb{E}\|\eta(t_0)\|_H^2 < \infty$, we have that

$$\mathbb{E}\|u(t) - \eta(t)\|_H^2 \leq K \exp(-\gamma(t - t_0)) \mathbb{E}\|u(t_0) - \eta(t_0)\|_H^2$$

for any $t \geq t_0$.

This property can be used to show the existence and uniqueness of an invariant measure. In [30], they show that the solution to

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - Au(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in G, t > 0 \\ u(0, \cdot) = u_0(\cdot) \end{cases}$$

where

- G is the upper half plane
- $\rho(x) = \exp(-|x|^2)$
- $Au := \frac{1}{\rho} \operatorname{div}(\rho \nabla u)$
- $u_0 \in L_\rho^2$
- Both f and b are Lipschitz in the second variable with LIP constant independent of x and $f(x, 0)$ and $b(x, 0)$ are in L^∞

has this exponential stability property and then refer to two propositions (11.2 and 11.4) in [20] which give the existence and uniqueness of an invariant measure.

In general, if we are given SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - Au(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in G, t > t_0 \\ u(0, \cdot) = u_0(\cdot) \end{cases}$$

with solution $u(t, t_0, u_0)$ that satisfies the exponential stability property and in addition satisfies the semigroup property (like the solution to the heat equation does), then we can show that for any t_0 , that $u(t, t_0, u_{t_0}) \rightarrow \mu$ as $t \rightarrow \infty$ where μ is invariant and $\mu = \mathcal{L}(u(t_0))$.

1.3 Case when $f \neq 0$

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

[2] proves the exist of an invariant measure when the initial condition is a function $\varphi_0(\cdot) \in L_\rho^p(\mathbb{R}^d)$ where the $p = 2 \wedge \nu$ and ν depends on the function f . I have not read this paper yet but I plan on it so I am not sure how they prove this.

I have tried to see if the proof from our paper (which is a mimic of the proof by Tessitore in [39]) would work but I ran into the following issue. Our proof for the existence of an invariant

measure relies on the the fact that for fixed $0 < \delta_0 \leq 1$ we can find in this set $\mathcal{K}(\Theta) \subset L_\rho^2$ defined in the following way

$$\mathcal{K}(\Theta) = \{(G(\delta_0, \cdot) * y(\cdot))(x) + (F_\alpha^{\delta_0} h)(x), \quad \text{where } |y|_{\hat{\rho}} \leq \Theta \text{ and } |h|_{L^q(0,1,L_\rho^2)} \leq \Theta\}$$

and if we still assume that $f \equiv 0$ then we can use the factorization theorem and say that the solution u can be written as

$$u(t_0, x) = (G(t_0, \cdot) * \varphi(\cdot))(x) + \frac{\sin(\alpha\pi)}{\pi} F_\alpha^{t_0}(Y(s, \cdot))(x)$$

and because of the compactness of the operators G and $F_\alpha^{t_0}$, the set $\mathcal{K}(\Theta)$ is relatively compact.

The issue when we now consider $f \not\equiv 0$ is the following. By the factorization theorem, we could now represent u as

$$u(t_0, x) = (G(t_0, \cdot) * \varphi(\cdot))(x) + \left(G(\circ, \cdot) \star f(\cdot, u(\circ, \cdot)) \right)(t_0, x) + \frac{\sin(\alpha\pi)}{\pi} F_\alpha^{t_0}(Y(s, \cdot))(x)$$

and we could try to mimic our proof and define a new $\mathcal{K}^*(\Theta)$ as follows

$$\mathcal{K}^*(\Theta) = \left\{ (G(\delta_0, \cdot) * y(\cdot))(x) + \left(G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x) + (F_\alpha^{\delta_0} h)(x), \right. \\ \left. \text{where } |y|_{\hat{\rho}} \leq \Theta \text{ and } |h|_{L^q(0,1,L_\rho^2)} \leq \Theta \text{ and } q \text{ has some property} \right\}$$

We need this new set $\mathcal{K}^*(\Theta)$ to be relatively compact in L_ρ^2 but then we would need the operator $\left(G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x)$ to be compact so that $\mathcal{K}^*(\Theta)$ remains relatively compact. So we need to see what space this function q needs to be in. Note that we want $f(x, u(t, x))$ to be a specific example of $f(x, q(t, x))$. Maybe we want the following

1. $t \mapsto q(x, t)$ is continuous or even just bounded on $t \in [0, T]$ for any $T > 0$.
2. $x \mapsto q(x, t)$ is in L_ρ^2

and the solution u satisfies these properties. However I was unable to show that the operator $\left(G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(t_0, x)$ was bounded (continuous). Following the steps in [39] we would first want to show something like

$$\left| \left(G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x) \right|_\rho^2 \leq K |q(\delta_0, \cdot)|_\rho^2$$

However the best I could do was show that

$$\left| \left(G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x) \right|_\rho^2 \leq K(T) |q(t^*, \cdot)|_\rho^2$$

where

$$q(t, x) \leq q(t^*, x) \quad \text{where we need to assume that } q(\cdot, x) \text{ has a max at } t^* \text{ on the interval } [0, T]$$

Things seem off about this so I will read Assing and Manthey's paper [2] to see how they consider the case when $f \not\equiv 0$.

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